

The Science of Climate Change

Elements of the climate system

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Today's topics

- The atmospheric circulation
- The ocean circulation
- The cryosphere
- The water cycle

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The five spheres



The atmospheric circulation

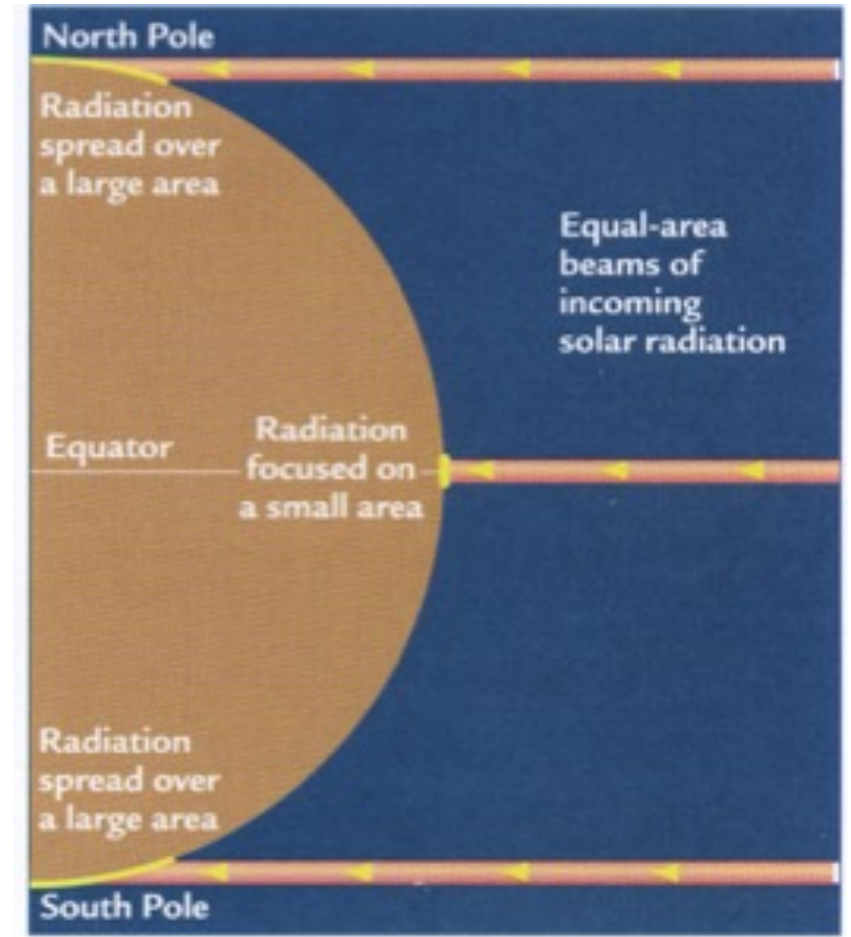
- **Q: Why does the atmosphere circulate?**

The atmospheric circulation

- **Q: Why does the atmosphere circulate?**
- The atmospheric circulation is fuelled by solar energy
- It does work to redistribute heat, both horizontally and vertically

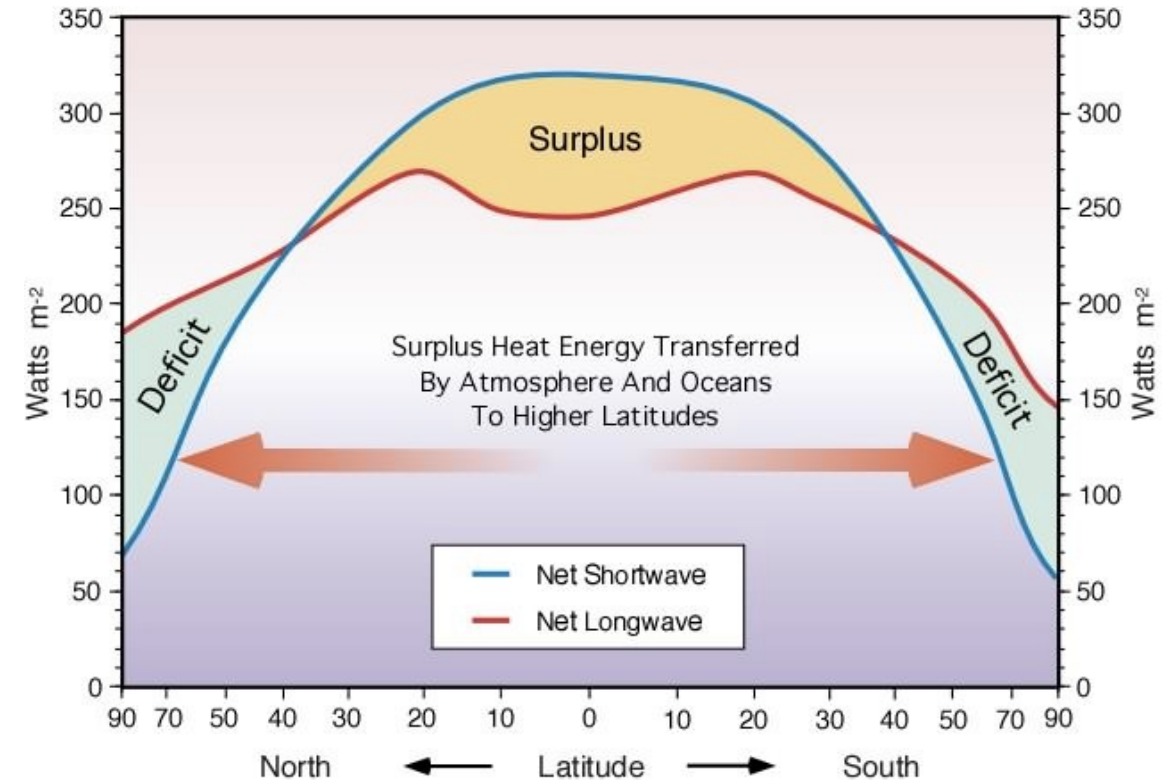
Meridional distribution of radiative heating

- So far we've looked only at global-mean radiative fluxes
- But on average, the tropics receive much more solar radiation than the high latitudes



Meridional distribution of radiative heating

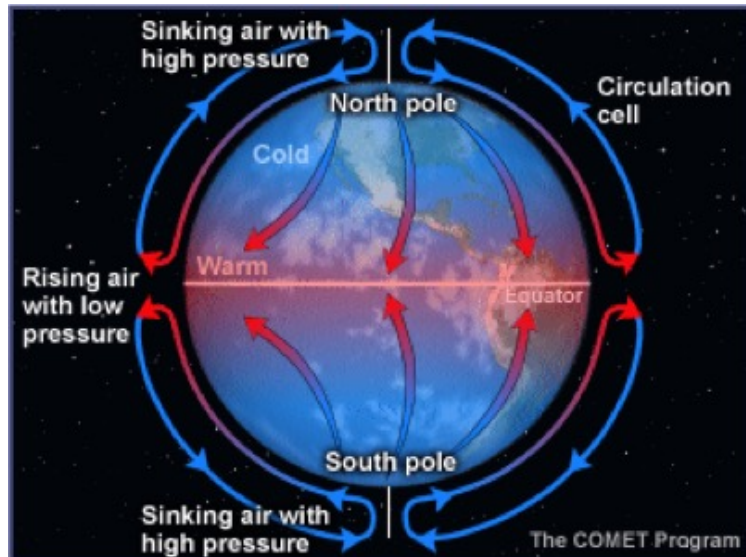
- Consequence: in the tropics, the atmosphere receives more heat than it can radiate away
- The excess radiative energy in the tropics is transported to higher latitudes by the circulation of the atmosphere and ocean
- Of course this also varies during the seasons
- Reminder:
 - Net shortwave = net solar radiative heating
 - Net longwave = net infrared radiative cooling



Mean meridional circulation

- As the air is heated from the surface, it becomes less dense than the air above it and starts to rise. This is called convection
 - The rising air will stop rising when it is cooled sufficiently by radiation
 - The rising air has to be replenished from the surface, and this creates cells

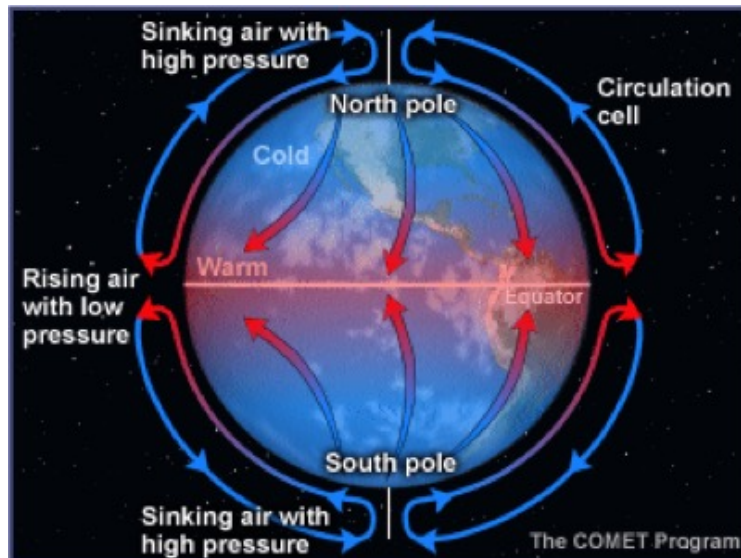
Atmospheric circulation without rotation



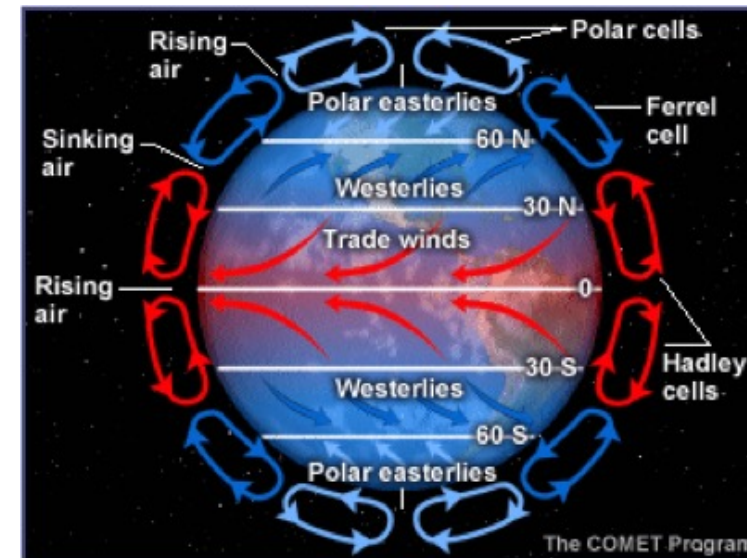
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- Due to the rotation of the earth, the cells break up

Atmospheric circulation without rotation

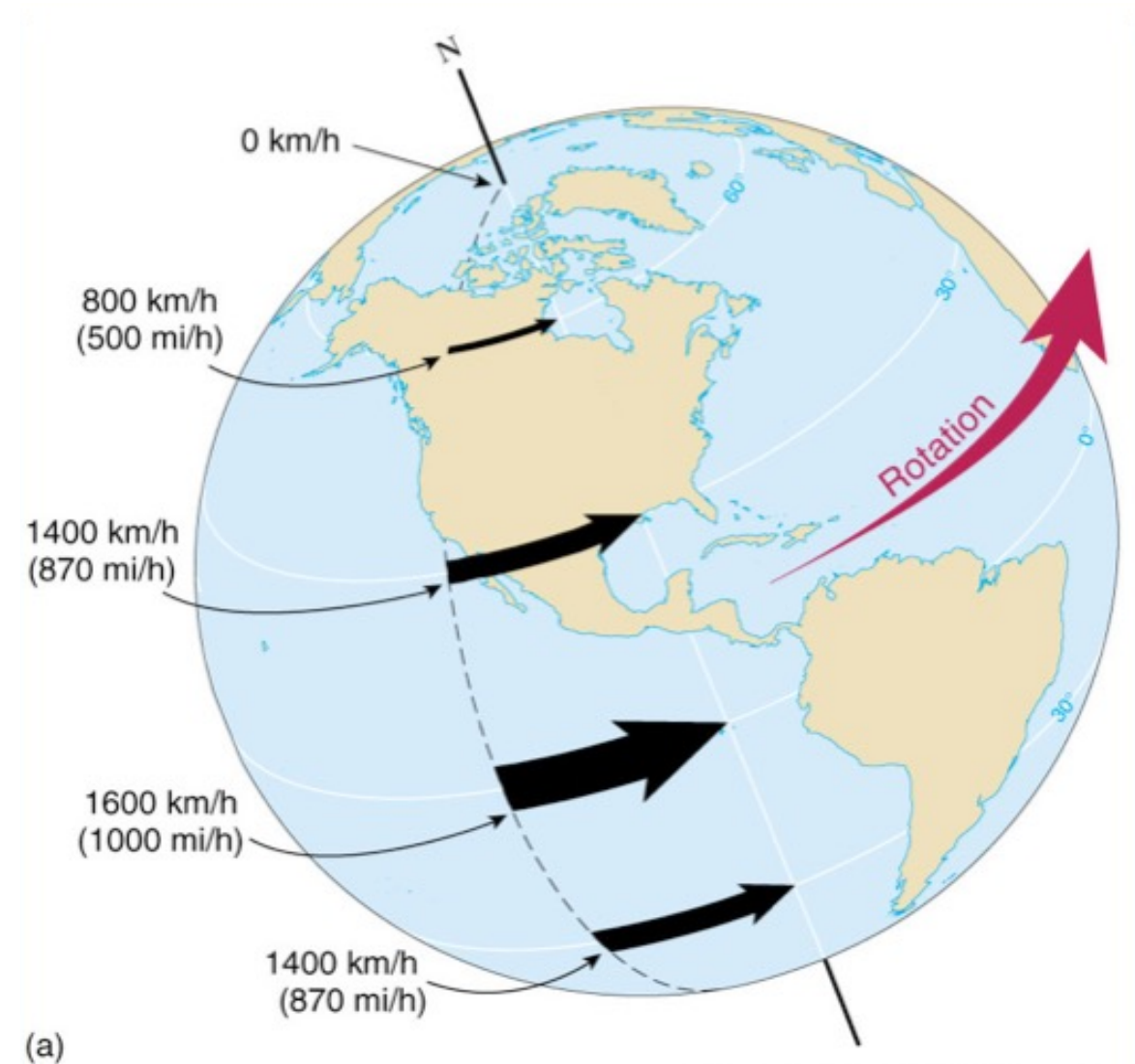


Atmospheric circulation with rotation



The effect of rotation

- The Coriolis force affects flow in a rotating frame of reference
- The earth is a sphere, rotating around its axis in 24 hours
- However, the distance travelled in one rotation depends on latitude
- This means that the velocity with which the earth surface rotates also varies with latitude
- This has great implications for the atmosphere and ocean circulation

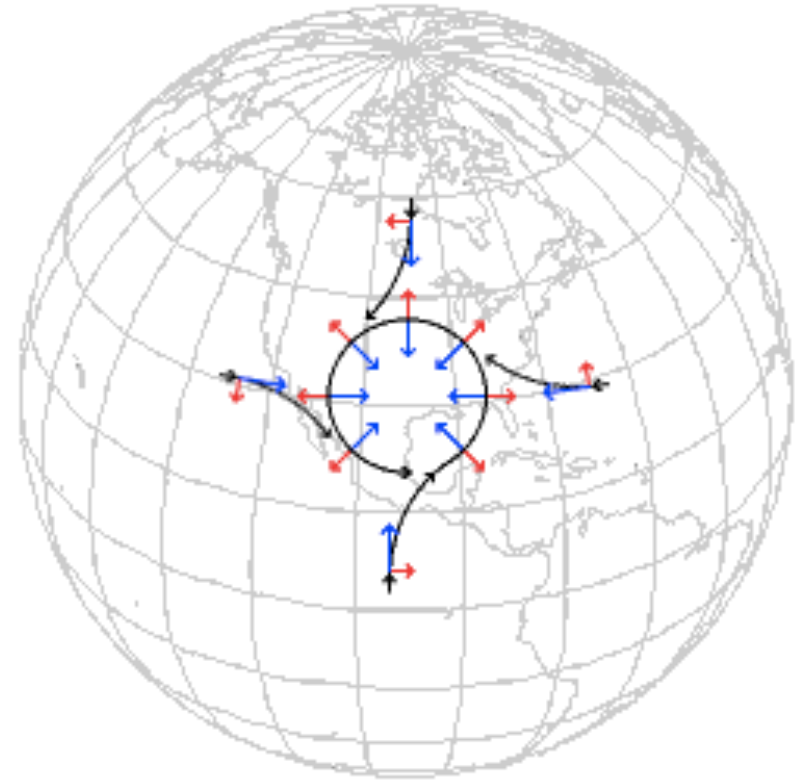


The Coriolis effect



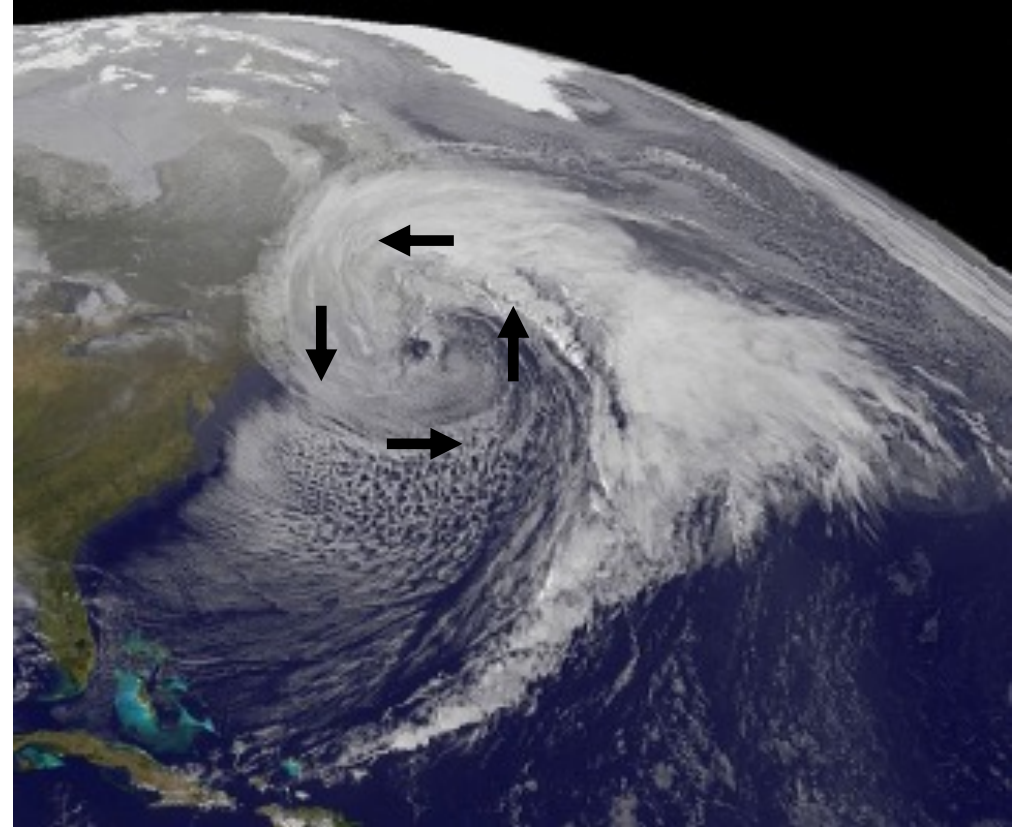
The Coriolis effect

- On Earth, the Coriolis effect results in trajectories being deflected to the right in the Northern Hemisphere (to the left in the Southern Hemisphere)



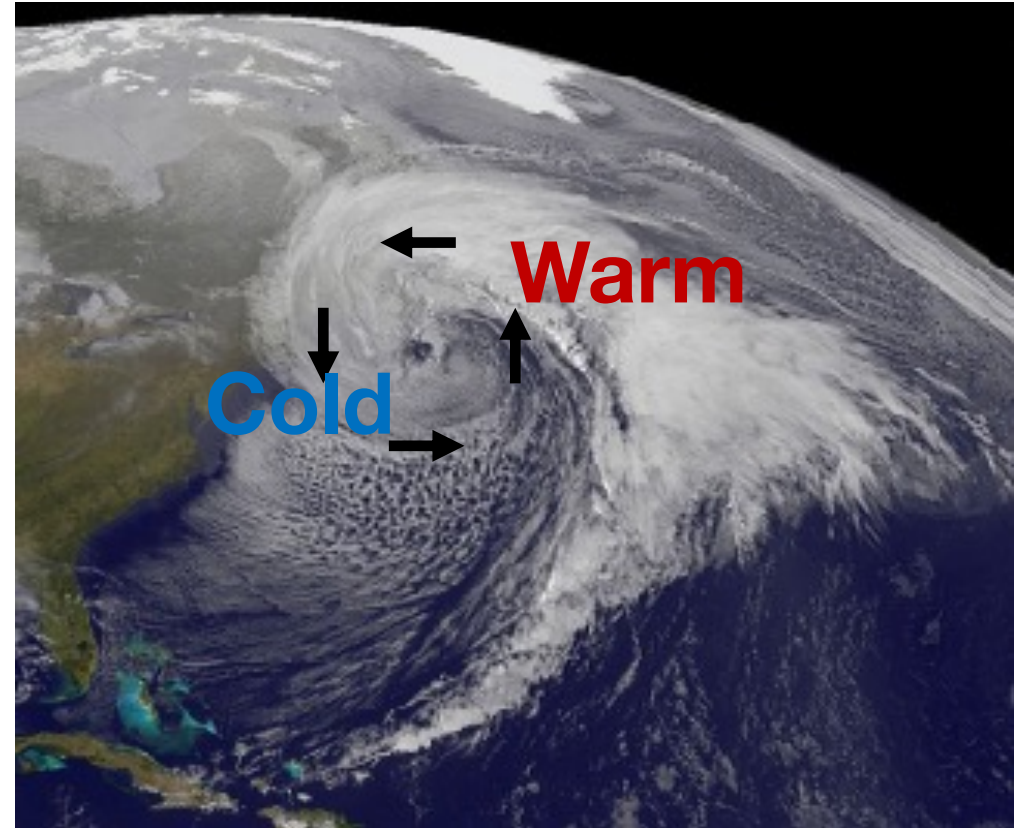
Weather systems (“eddyies”)

- [Map of current surface winds](#)
- The Coriolis effect is responsible for the counter-clockwise rotation of storms (clockwise in the Southern Hemisphere)



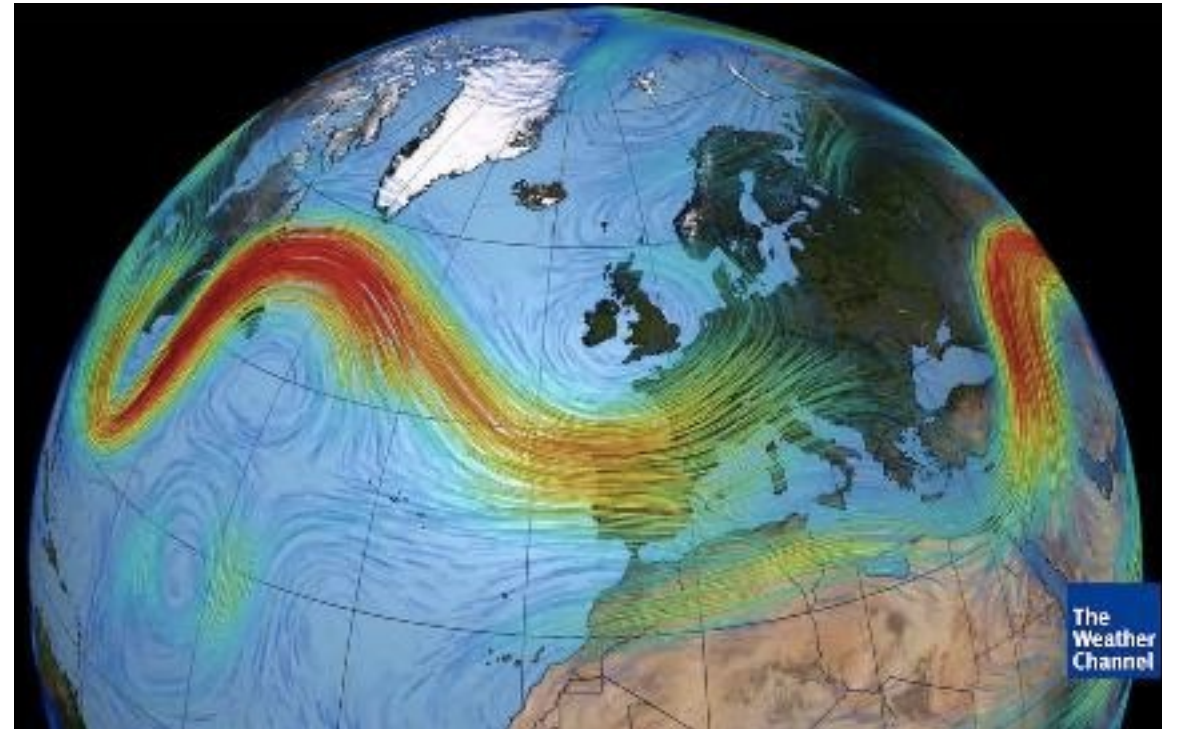
Weather systems (“eddyies”)

- [Map of current surface winds](#)
- The Coriolis effect is responsible for the counter-clockwise rotation of storms (clockwise in the Southern Hemisphere)
- These storms also make a large contribution to the meridional transport of heat in the atmosphere, by “mixing” cold and warm air
- They occur mainly in the middle latitudes: “mid-latitude storms”



The jet streams

- [Map of 250 hPa winds](#) (~10 km height)
- Belts of strong eastward winds in the middle latitudes of each hemisphere, in the upper atmosphere (~10 km)
- Coincident with strong temperature gradients: typically warm air is to the south, cold air to the north
- Both the jet streams and mid-latitude storms get their energy from the temperature difference between tropics and high latitudes



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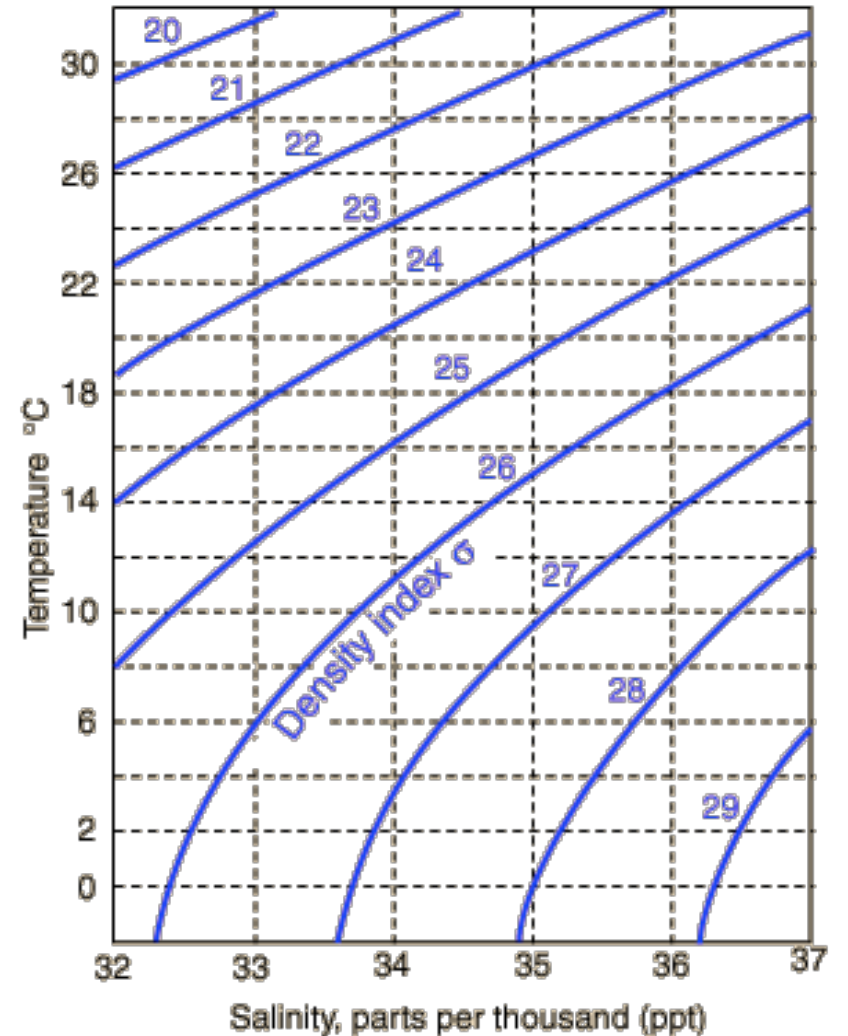
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The ocean circulation

- [Map of ocean surface currents](#)
- The Gulf stream is clearly visible
- Ocean currents are driven by two factors:
 - Surface winds
 - Temperature and salinity differences (both of which control density)

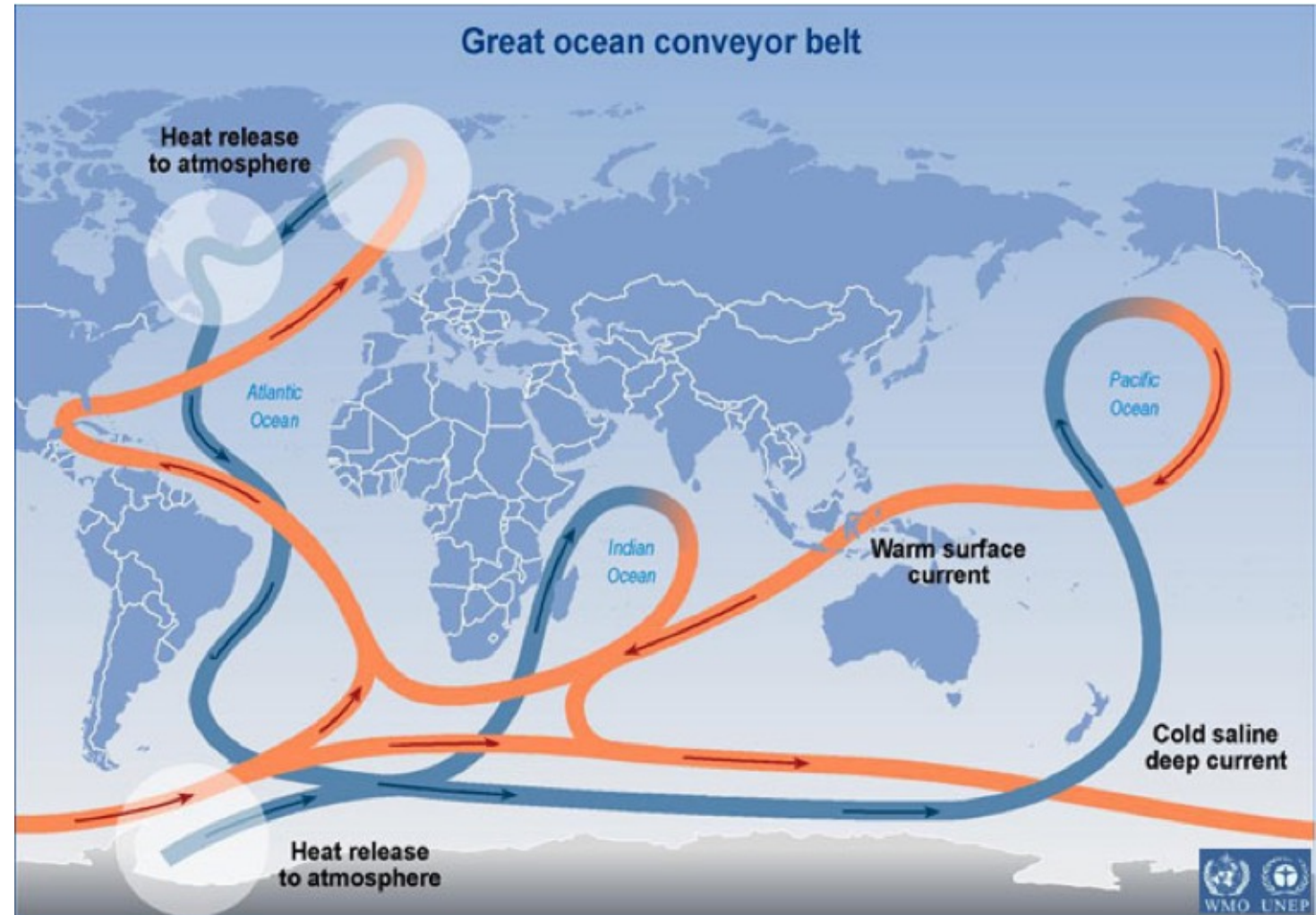
Density of sea water

- Density increases with **decreasing temperature** and **increasing salinity**
- The densest waters are found near the sea ice margins: very cold water + high salinity, because salt is left behind when ice forms

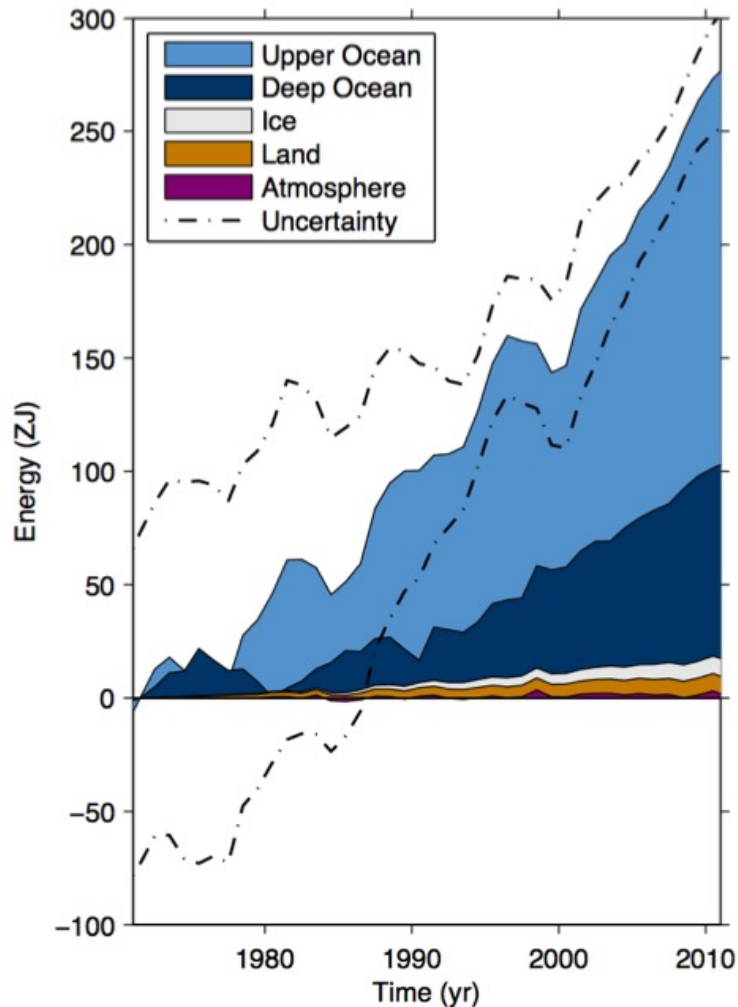


Mean circulation and heat transport in the ocean

- Ocean circulation also transports heat from low to high latitudes, on average
- Much of the meridional transport is in the Atlantic: “Atlantic meridional overturning circulation” (AMOC)
- Much more complicated pattern than in the atmosphere
- This picture is oversimplified! (but still useful)



Ocean heat storage



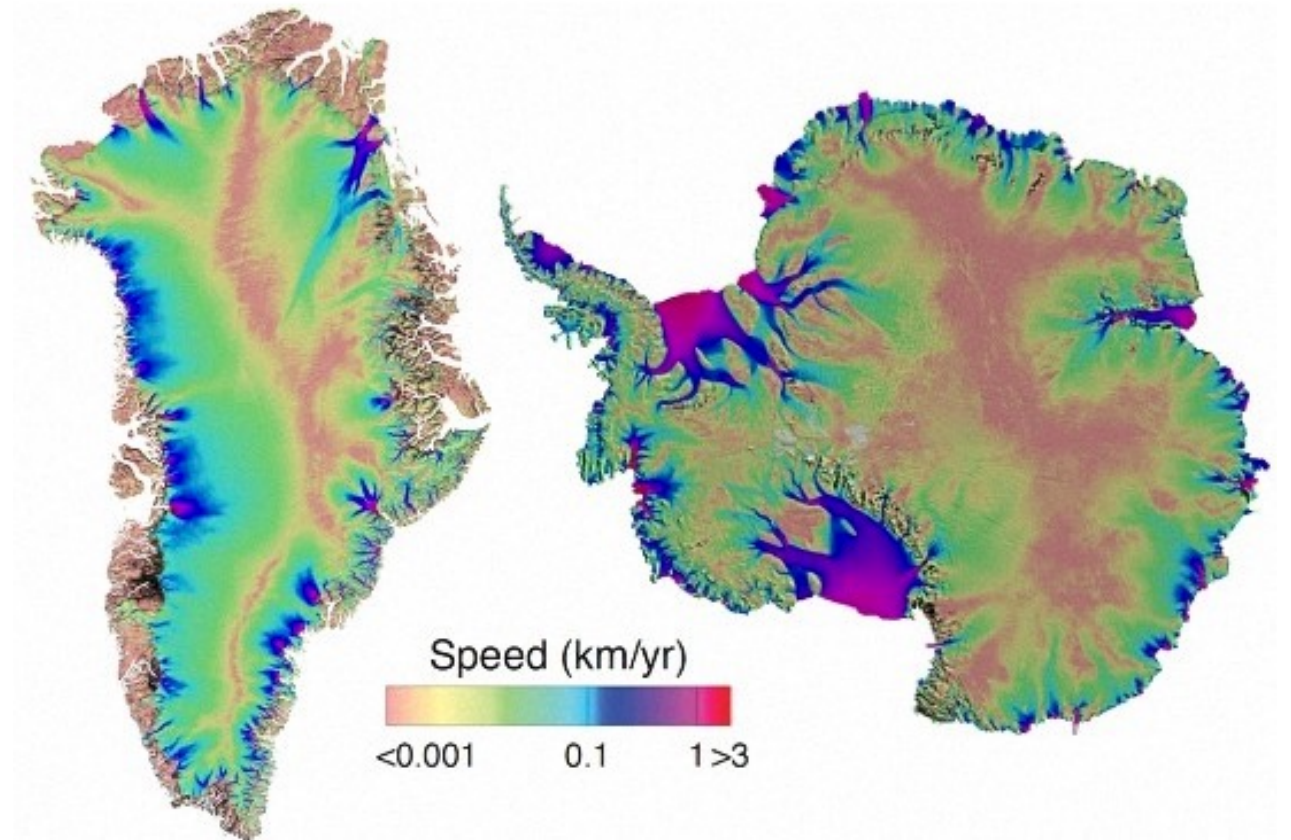
- The vast majority of the extra heat stored in the climate system goes into the ocean (about 90%)
- The oceans are deep and water has a high **heat capacity**, meaning it takes a lot of energy to heat water by a certain amount
- By comparison, air and land have a small heat capacity
- This means that the oceans take a very long time to come to equilibrium. Even if we stopped emitting GHG today, the oceans would continue warming for centuries

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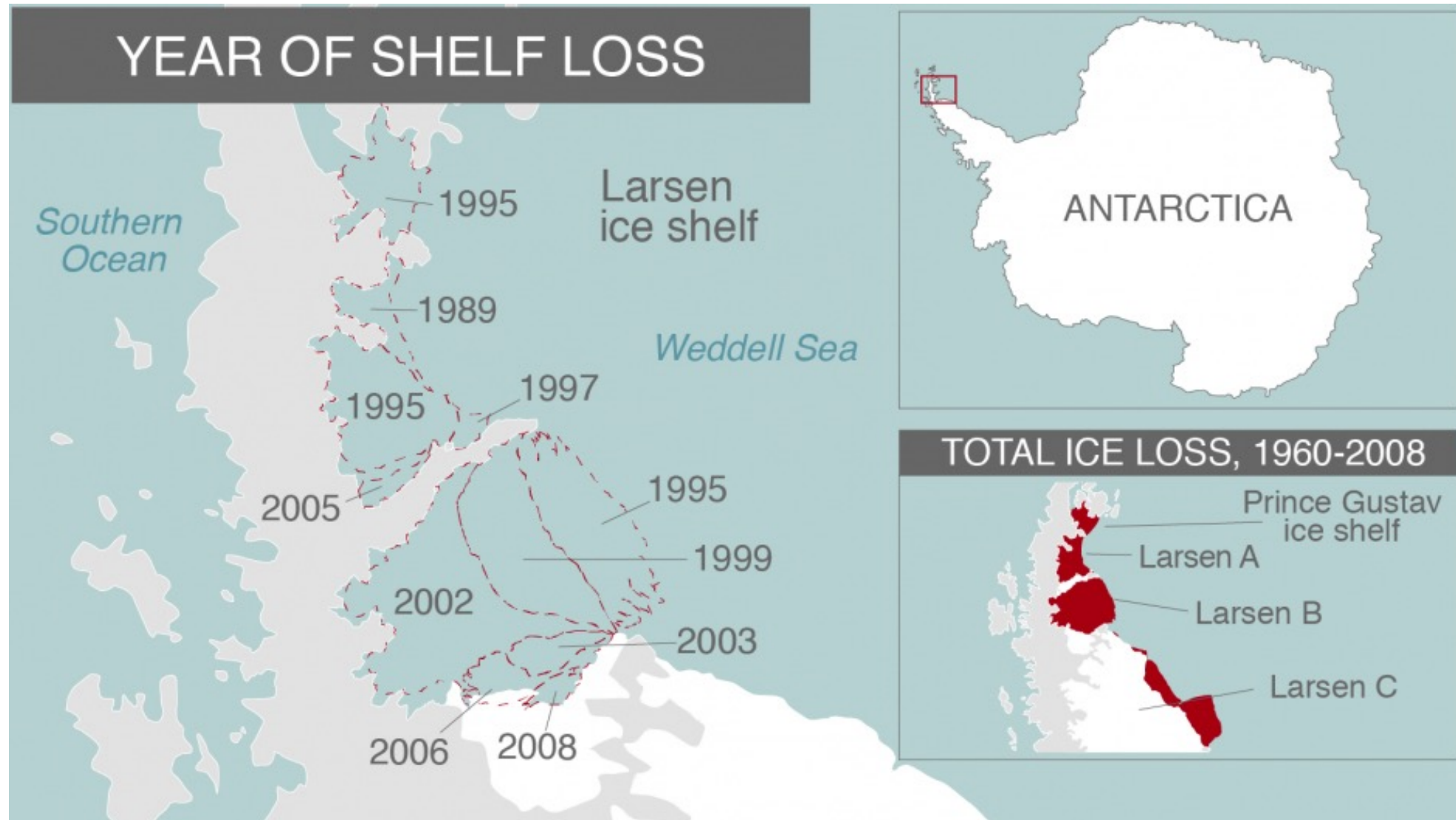
The cryosphere: ice sheets and ice shelves

- Ice sheet: large mass of continental ice
- Ice shelf: a thick floating platform of ice that forms where a glacier or ice sheet flows onto the sea
- On long time scales, ice sheets flow



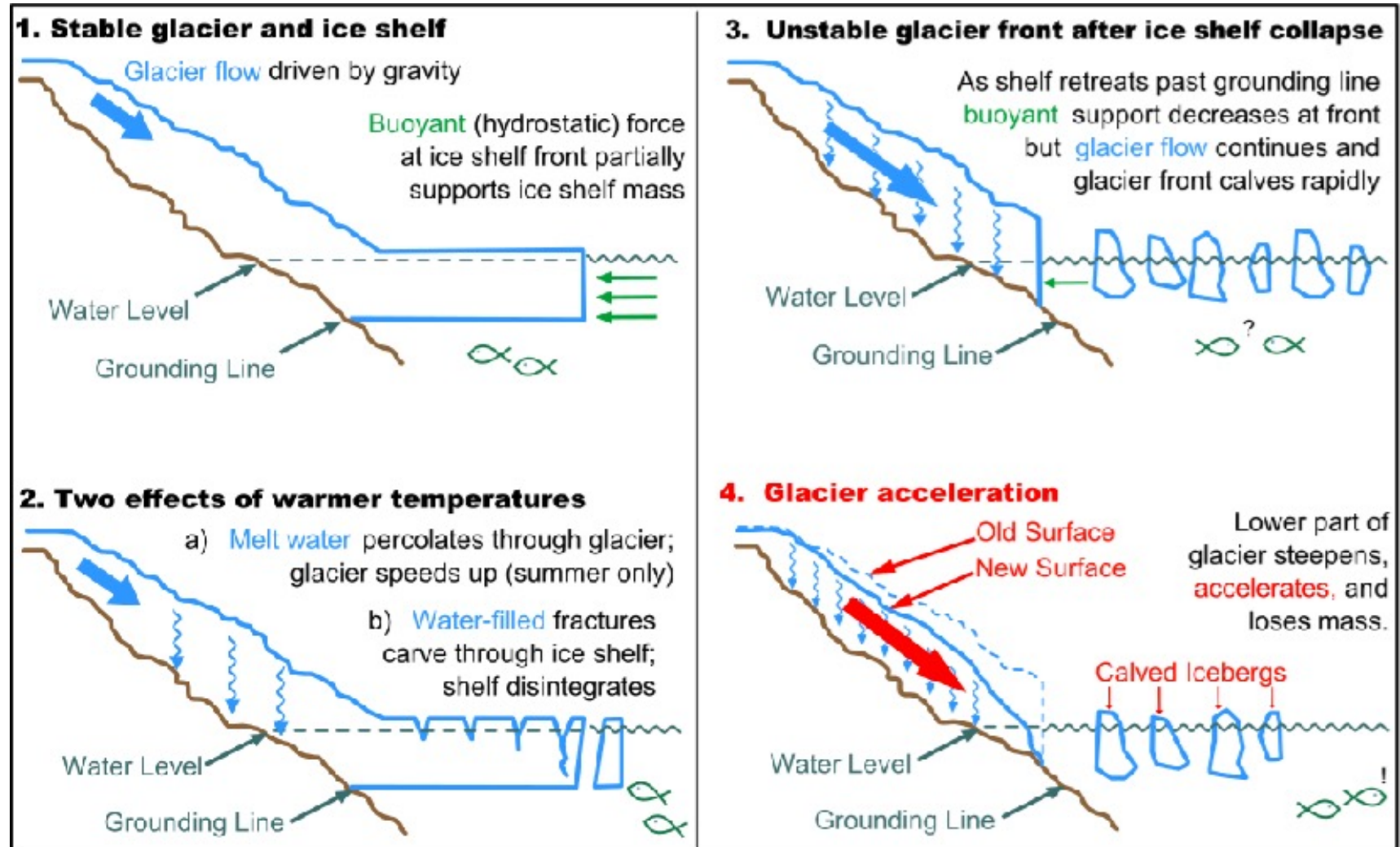
Mouginot et al, Remote Sensing, 2017

The Larsen ice shelves



Ice shelves as buttresses for glaciers

- Key point is that ice sheets can become unstable when ice shelves disappear
- Potential for accelerated sea level rise if that happens

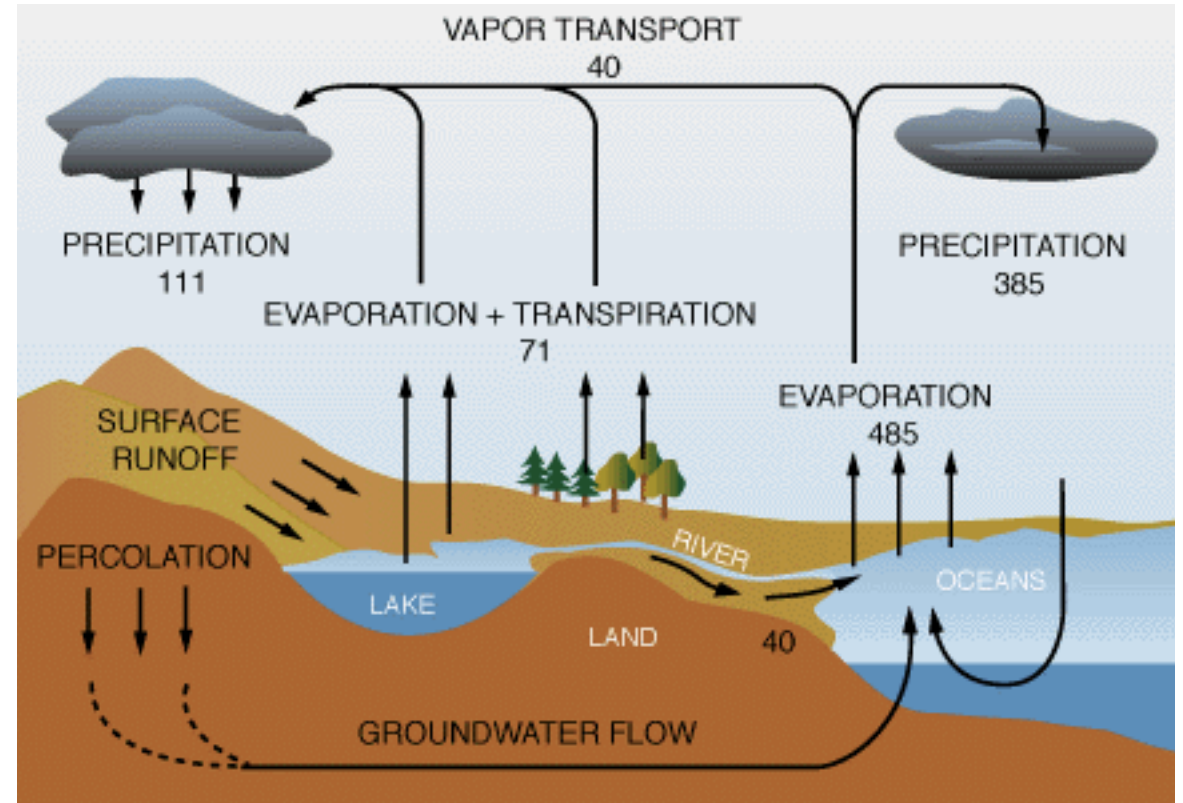


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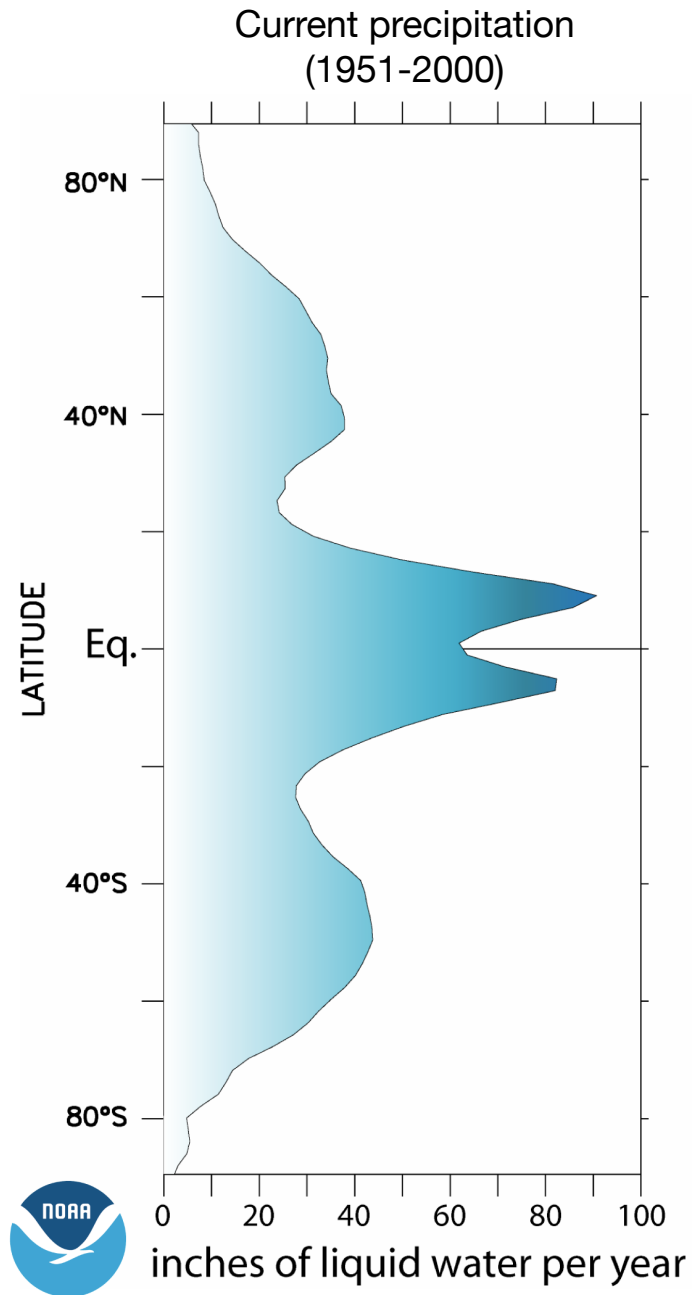
The water cycle

- Water is transported from the oceans to the continents as vapour
- It flows back via runoff (surface and groundwater)
- **Q: Where on Earth is it driest? And where is it wettest?**



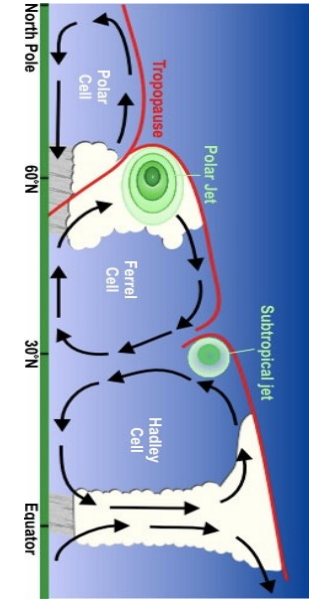
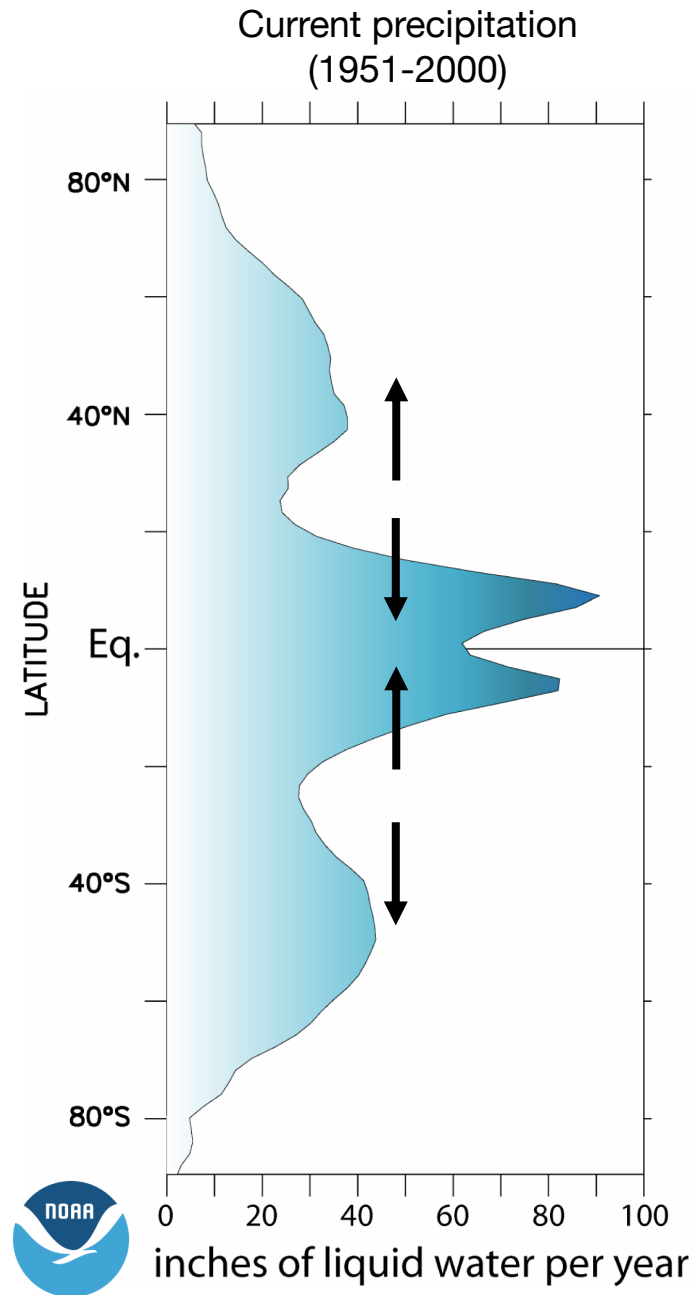
The water cycle

- Some regions receive much more precipitation than others
- Dry regions are typically in the subtropics, about 20 to 40 degrees latitude, including deserts like the Sahara



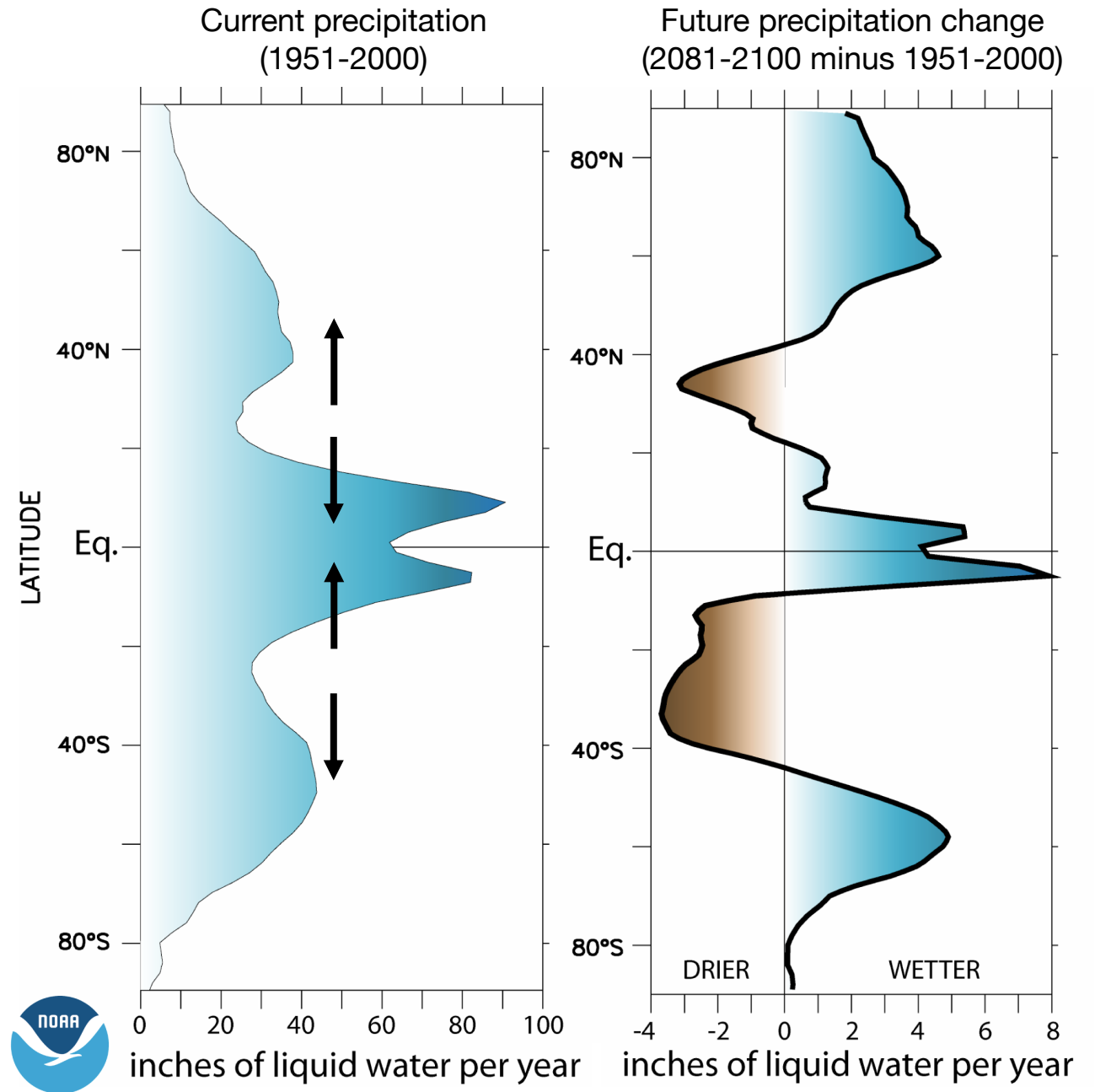
The water cycle

- Some regions receive much more precipitation than others
- Dry regions are typically in the subtropics, about 20 to 40 degrees latitude, including deserts like the Sahara
- The distribution of wet and dry regions is determined by the circulation of the atmosphere, which transports heat and moisture



Future changes

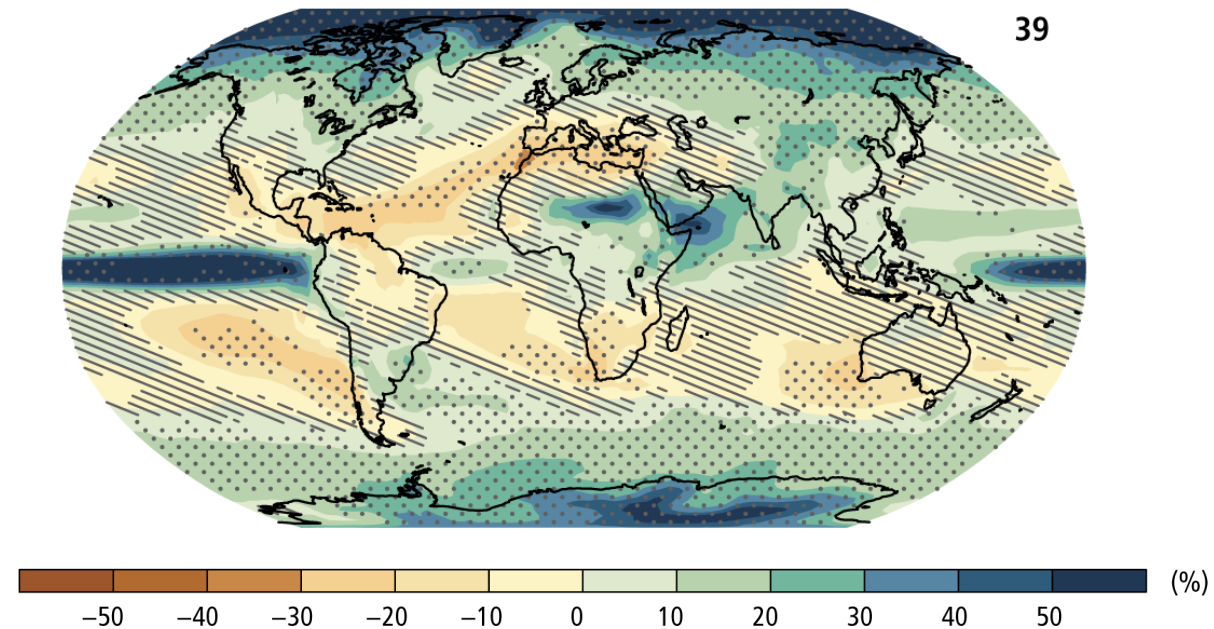
- On average, the water cycle will intensify with global warming:
“wet gets wetter, dry gets drier”
- This is a direct consequence of higher moisture content in the atmosphere (Clausius-Clapeyron, lecture 4)
- Important for future water management in subtropical/Mediterranean regions



Future changes

- Precipitation is projected to decrease over the Mediterranean, South Africa, Australia
- Generally more precipitation in Equatorial regions and at high latitudes
- Stippling where 90% of models agree on sign of change
- Hatching where the change is uncertain

Change in precipitation in RCP8.5 scenario
(2081-2100 minus 1986-2005)



IPCC AR5 WG1 SPM

Recap: Elements of the climate system

- The climate is controlled by all components of the earth system: a complicated and interlocked set of five “spheres”
- The spheres control heat transport (through winds and ocean currents), as well as the cycles of water and carbon
- The ocean has taken up approximately 90% of the heat produced by human activities
- A warmer atmosphere can hold more moisture, but this doesn’t lead to more rainfall everywhere.
 - Generally, wet regions will get wetter, already dry regions will get even drier

Further reading

- Global warming: Understanding the forecast (D. Archer), chapters 8-10
- [IPCC AR6 WG1](#) chapter 9 (ocean and cryosphere changes)
- [What climate models tell us about future rainfall](#) (Carbon Brief, Jan 2018)